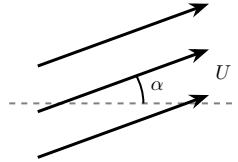
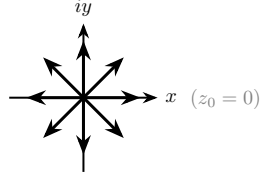
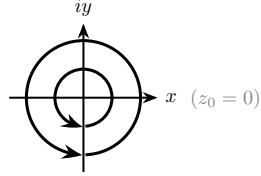
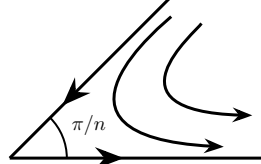
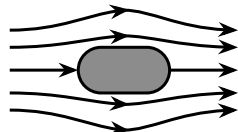
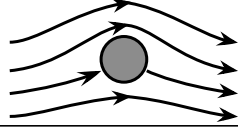


A Brief Handbook of Elementary Ideal Flows¹

Elementary flow	Complex potential $F = \phi + i\psi$	Potential $\phi(x, y)$	Streamfunction $\psi(x, y)$	Sketch of streamlines
①. Uniform flow in the direction α	Az with $A = Ue^{-i\alpha}$ where U is real but A is complex	$U(x \cos \alpha + y \sin \alpha)$ or $Ur \cos(\theta - \alpha)$	$U(y \cos \alpha - x \sin \alpha)$ or $Ur \sin(\theta - \alpha)$	
②. Source ($Q > 0$; for sink replace Q by $-Q$) located at $z_0 = x_0 + iy_0$	$\frac{Q}{2\pi} \log(z - z_0)$	$\frac{Q}{4\pi} \ln[(x - x_0)^2 + (y - y_0)^2]$ or, for $z_0 = 0$, $\frac{Q}{2\pi} \ln r$	$\frac{Q}{2\pi} \tan^{-1} \left(\frac{y - y_0}{x - x_0} \right)$ or, for $z_0 = 0$, $\frac{Q}{2\pi} \theta$	
③. Vortex located at $z_0 = x_0 + iy_0$	$\frac{-i\Gamma}{2\pi} \log(z - z_0)$	$\frac{\Gamma}{2\pi} \tan^{-1} \left(\frac{y - y_0}{x - x_0} \right)$ or, for $z_0 = 0$, $\frac{\Gamma}{2\pi} \theta$	$-\frac{\Gamma}{4\pi} \ln[(x - x_0)^2 + (y - y_0)^2]$ or, for $z_0 = 0$, $-\frac{\Gamma}{2\pi} \ln r$	
④. Doublet located at $z_0 = x_0 + iy_0$ and rotated by angle α	$\frac{me^{i\alpha}}{z - z_0}$	$\frac{m \cos \left(\alpha - \tan^{-1} \left(\frac{y - y_0}{x - x_0} \right) \right)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}$ or, for $z_0 = 0$, $\frac{m \cos(\alpha - \theta)}{r}$	$\frac{m \sin \left(\alpha - \tan^{-1} \left(\frac{y - y_0}{x - x_0} \right) \right)}{\sqrt{(x - x_0)^2 + (y - y_0)^2}}$ or, for $z_0 = 0$, $\frac{m \sin(\alpha - \theta)}{r}$	
⑤. Flow in a corner ² of angle π/n	Az^n	... or $Ur^n \cos(n\theta)$	... or $Ur^n \sin(n\theta)$	
⑥. Flow around a Rankine body (a is a real number/horizontal shift) ①{ $\alpha = 0$ } + ②{ $z_0 = -a$ } - ②{ $z_0 = a$ }	$Uz + \frac{Q}{2\pi} \log \left(\frac{z + a}{z - a} \right)$	$Ux + \frac{Q}{4\pi} \ln \left[\frac{(x + a)^2 + y^2}{(x - a)^2 + y^2} \right]$...	$Uy + \frac{Q}{2\pi} \left[\tan^{-1} \left(\frac{y}{x + a} \right) - \tan^{-1} \left(\frac{y}{x - a} \right) \right]$...	
⑦. Flow around a cylinder with circulation ①{ $\alpha = 0$ } + ④{ $\alpha = 0, m = UR^2, z_0 = 0$ } - ③{ $z_0 = 0$ }	$U \left(z + \frac{R^2}{z} \right) + \frac{i\Gamma}{2\pi} \log z$	... $U \left(1 + \frac{R^2}{r^2} \right) r \cos \theta - \frac{\Gamma}{2\pi} \theta$	... $U \left(1 - \frac{R^2}{r^2} \right) r \sin \theta + \frac{\Gamma}{2\pi} \ln r$	

¹Inspired by Tables 12.1 and 12.2 of Granger's *Fluid Mechanics* (Dover Publications, 2012). Flow 1–5 are the “building blocks” and the rest are superposition examples.²The case $n = 2$ corresponds to stagnation-point flow onto a horizontal plate.